# Wheeled Mobile Robot Autonomy Under Degradation: A Survey of Methods and Evaluation

***Abstract*—Wheeled mobile robots face changing sensing, terrain, map, communication, and computational conditions and errors of learned inference, which violate the assumptions of navigation algorithms. Reliable navigation requires estimation, planning, and control to account for these changes jointly, since errors and delays in one component affect decisions elsewhere. This survey compares 245 publications from January 2021 to November 2026 on localization and mapping, traversability and motion control, semantic navigation, multi-robot coordination, and fault management. It codes each work by the degradation sources it addresses, the assumptions they violate, and the evidence linking its response to navigation outcomes. Selective sensing and fusion, adaptive motion models, map revision, and constrained action selection are complementary responses whose effectiveness depends on valid state estimates, fresh information, and feasible actions. Evaluation practice is compared by measurement level, test conditions, and the events underlying risk or diagnostic claims. In this purposively selected sample, 81 of 244 technical publications (33%) report a task outcome under a named degradation condition in the same experiment, from 1 of 67 localization studies to 36 of 56 planning studies. Only 27 (11%) inject or vary that condition against a comparator. The survey's agenda for joint evaluation covers four interfaces: localization-triggered recovery, semantic reasoning with fallback under resource limits, terrain decisions under pose uncertainty, and accepting shared data after reconnection. Evaluating them requires linking degradation onset, detection, and executed response to safety and task performance under matched conditions. Further challenges concern platform transfer, computational resources, sustained operation, and assurance under changing information quality.**

***Index Terms*—Wheeled mobile robots, robust navigation, localization, motion planning, multi-robot systems, fault tolerance, performance evaluation.**

## I. Introduction

AUTONOMOUS wheeled robots transport goods, inspect facilities, explore unfamiliar sites, and assist people indoors and outdoors. To navigate, a robot must estimate its pose from imperfect measurements, represent obstacles and traversable space, plan a route that its chassis can execute, and track that route as new observations arrive, while a team member must also reconcile its own observations with information received from teammates. Reliable operation therefore depends both on the individual algorithms and on the information that passes between estimation, mapping, planning, and control.

Degradation occurs when operating conditions depart from the assumptions of these algorithms. Ground-robot datasets document combinations of darkness, lift transitions, abrupt braking, and uneven terrain [1], [2]. Beyond these sensing conditions, long-term deployments depend on charging, network availability, and operator assistance [3]. Addressing these conditions requires methods that relate the quality of the available information to the actions that a robot can still execute safely.

Geometric, learned, and language-based methods address these requirements in combination. Representative simultaneous localization and mapping (SLAM) systems [4], [5] and the dynamic-window [6] and timed-elastic-band [7] planners that recur as comparators [8], [9] illustrate how these components assume different inputs and fail in different ways. Existing surveys examine individual aspects of the problem. They cover SLAM in extreme subterranean environments [10], global LiDAR localization [11], millimeter-wave radar in robotics [12], automated-vehicle perception under adverse weather [13], terrain traversability [14] and its risk assessment [15], safe learning in robotics [16], learned navigation policies [17], social navigation [18], vision-and-language navigation [19], collaborative SLAM [20], and foundation models in robotics [21]. Each centers on one function, sensor, task setting, or method family, although the foundation-model survey spans perception, decision-making, and control. The review of learned navigation policies also serves as the context review of the coded corpus. These surveys were found through targeted searches for each navigation function and through suggestions of an internal review of an earlier draft, not through a systematic search. Table I compares the scopes that their abstracts state.

Automated driving has established a vocabulary for operation under degradation. SAE J3016 defines levels of driving automation through the roles of the user and the system in the dynamic driving task and its fallback [22]. ISO 21448 concerns the safety of the intended functionality, namely hazards that arise from insufficiencies of the specification or of sensing and processing performance rather than from faults, for series-production road vehicles [23]. Runtime monitors and fallback maneuvers turn these notions into checks and actions during operation. A check of the operational design domain, the conditions that an automated function is designed for, can flag laser scans unsuitable for matching before localization runs [24]. A minimum risk maneuver, in the terms of the cited study, brings an automated vehicle to a minimum risk condition after an unanswered takeover request or a severe system failure [25]. A stated operating domain, a monitored validity condition, and a fallback such as stopping in place transfer directly to wheeled robots. The road-vehicle setting does not cover unstructured terrain, robot teams, or semantic task interpretation. On a slope or in a crowd, stopping in place may itself leave a wheeled robot at risk.

This survey organizes the literature around three questions: which navigation assumption a degradation affects, how a method changes the information or action available to the robot, and what evidence connects that change to navigation performance. A codebook applies the first and third questions to every corpus work. Fig. 1 crosses the nine coded degrada-

TABLE I
RELATED SURVEYS CHARACTERIZED FROM THEIR ABSTRACTS. "NOT STATED" MEANS THAT THE ABSTRACT DOES NOT NAME THE TOPIC, WHICH DOES NOT EXCLUDE COVERAGE IN THE BODY OF THE SURVEY.

| Survey | Platform scope | Degradation coverage | Recovery or faults | Evaluation practice |
|---|---|---|---|---|
| Ebadi et al. (2024) [10] | Ground and aerial robots | Obscurants, tight computation | Not stated | Six SubT teams, datasets |
| Yin et al. (2024) [11] | Single and multiple robots | Not stated | Not stated | Not stated |
| Harlow et al. (2024) [12] | Robots using radar | Fog, dust, smoke | Not stated | Datasets, calibration |
| Zhang et al. (2023) [13] | Automated road vehicles | Weather effects on sensors | Not stated | Datasets, simulators, chambers |
| Sevastopoulos and Konstantopoulos (2022) [14] | Mobile robots | Slope, roughness, surface | Not stated | Not stated |
| Benrabah et al. (2024) [15] | Ground vehicles, robots | Traversability risk | Not stated | Risk metric consistency |
| Brunke et al. (2022) [16] | Robots, learned control | Uncertain dynamics | Not stated | Physics-based benchmarks |
| Zhu and Zhang (2021) [17] | Mobile robots | Not stated | Not stated | Not stated |
| Mavrogiannis et al. (2023) [18] | Robots in public crowds | Not stated | Not stated | Evaluation methods |
| Gu et al. (2022) [19] | Embodied language agents | Not stated | Not stated | Evaluation metrics |
| Lajoie et al. (2022) [20] | Multi-robot systems | Communication, resources | Not stated | Not stated |
| Firoozi et al. (2025) [21] | Not stated | Not stated | Not stated | Not stated |
| This survey (2026) | Wheeled robots, transfer cases | Nine coded sources | Monitoring, recovery | Coded evidence, sensitivity |

tion sources, grouped by the six assumptions they violate, with the five navigation functions of Sections IV–VIII. As its first contribution, this survey compares responses to degradation across five navigation functions and their sensing, model, and action requirements, with reported drivetrain effects, distinguishing component improvements from executed recovery. Second, it codes the degradation sources and evaluation evidence of the 244 technical publications and finds that, in this purposively selected sample, 33% report a task outcome under a named degradation condition in the same experiment, from 1% of localization and mapping studies to 64% of planning and control studies. A language-model second coding of 40 sampled works checks these codes. Third, it proposes four interfaces between existing methods for joint evaluation, locates them in a ranking of the coded function couplings, and outlines a matched comparison for each.

The remaining sections cover wheeled motion and the navigation stack (Section II), selection and coding (Section III), the technical synthesis (Sections IV–VIII), evaluation practice (Section IX), open challenges (Section X), and conclusions (Section XI).

## II. BACKGROUND

### A. Wheeled Platforms and Motion Constraints

A wheeled mobile robot moves through driven or steered rolling contacts whose footprint, arrangement, and contact conditions constrain the paths it can execute. Differential-drive, skid-steer, Ackermann, and omnidirectional platforms differ in how they turn and whether they can move sideways [26]. For a planar reference point with position $(x, y)$ and heading $\psi$, a common idealization is

$$\dot{x} = v\cos\psi, \qquad \dot{y} = v\sin\psi, \qquad \dot{\psi} = \omega, \tag{1}$$

where $v$ is longitudinal speed and $\omega$ is angular speed. Under pure rolling, a differential-drive robot with wheel radius $r$, track width $b$, and right and left wheel angular speeds $\omega_R$ and $\omega_L$ satisfies

$$v = \frac{r}{2}(\omega_R + \omega_L), \qquad \omega = \frac{r}{b}(\omega_R - \omega_L). \tag{2}$$

An ideal bicycle approximation for Ackermann steering instead gives

$$\omega = \frac{v}{L}\tan\delta, \tag{3}$$

where $L$ is the wheelbase and $\delta$ is the steering angle. Table II lists how slip, steering limits, and wheel arrangement violate (1)–(3) for each drivetrain and which recovery motions remain available. On uneven ground, suspension, pitch, roll, clearance, traction, and payload further affect feasibility. A planar collision-free path may therefore be physically unsuitable even when localization is accurate.

TABLE II
DRIVETRAIN TYPE, THE KINEMATIC ASSUMPTION THAT DEGRADATION VIOLATES, AND THE RECOVERY MOTIONS THAT REMAIN AVAILABLE.

| Drive | Degradation effect | Recovery motions |
|---|---|---|
| Differential, (1)–(2) | Slip and wheel lift break the encoder model. Covariance can grow along unexplained motion [27] | Turning in place, reversing. Sideways moves need rotation [28], [29] |
| Skid steer, (2) | Slip varies with speed, maneuver, and surface [30], [31]. A fixed model misses side slip at speed [32] | Turning in place at the cost of odometry. Reversing |
| Ackermann, (3) | Turning radius and input bounds limit corrections [33], [34] | No turning in place. Reorientation needs space or reversing |
| Omni-directional | Lateral motion from the wheel arrangement [26]. Mecanum rollers can slip on uneven ground [35]. Another Mecanum study reports no drive-specific effect [36] | Sidestepping and turning in place |

Except for the second Mecanum study, corpus citations mark works coded with a drivetrain-dependent effect. The corpus has 10 such works under the definition narrowed in codebook version 1.1, after a recheck of all works for missed cases (Section S7 of the supplementary material). A wheeled-legged robot also recovers by repositioning its wheels to reshape the support polygon, which conventional wheels cannot do [37].

| Violated assumption | Degradation source | Sec. IV Localization | Sec. V Planning | Sec. VI Semantic | Sec. VII Multi-robot | Sec. VIII Faults | Response families |
|---|---|---|---|---|---|---|---|
| A1. Measurements are informative and correctly associated | Perceptual degradation | 38/34/9 W18 R3 N17 | 3/1/1 W3 | 2/2/0 W2 | 4/3/0 W2 G1 N1 | 6/5/0 W4 R1 N1 | IV-A selective updates<br>IV-B complementary sensing<br>IV-E relocalization<br>VIII-B integrity monitors |
| | Sensor fault and false association | 6/5/3 W2 R1 G1 N2 | 1/0/0 W1 | 1/1/1 W1 | 8/6/3 W5 N3 | 10/10/6 W7 R3 | |
| A2. Wheels roll on supporting ground with known contact | Terrain and wheel-ground contact | 17/13/4 W13 R1 N3 | 20/20/2 W17 R1 G2 | 4/3/0 W4 | 3/2/0 W2 G1 | 5/5/1 W4 N1 | IV-C contact priors<br>V-A terrain risk<br>V-E tracking, delay |
| A3. The mapped scene is static or persistent | Moving agents | 16/13/0 W6 R2 N8 | 23/21/7 W21 G1 N1 | 8/8/1 W7 G1 | – | 8/7/0 W6 N2 | IV-D persistent maps<br>V-C prediction<br>V-D risk constraints<br>VI-A semantic memory |
| | Map and scene change | 16/16/3 W7 R2 N7 | 2/2/1 W2 | 7/6/4 W7 | 1/1/0 W1 | 4/0/0 W4 | |
| A4. Learned and semantic inference is valid | Learned and semantic inference error | 4/3/1 W1 R2 N1 | 14/12/1 W13 G1 | 7/5/0 W7 | – | 9/8/0 W8 N1 | VI-B learned actions<br>VI-C candidate selection<br>VI-D decision timing<br>VIII-B integrity monitors |
| A5. Shared information is timely and consistent | Communication loss, delay and bandwidth | 2/1/0 W1 R1 | 3/2/2 W3 | – | 30/18/11 W20 G3 N7 | 2/1/1 W2 | V-E tracking, delay<br>VII-B distributed estimation<br>VII-C rendezvous, repair |
| A6. Actuation, computation and energy remain available | Actuator fault and saturation | – | 5/3/3 W5 | 1/1/1 W1 | 5/3/1 W5 | 8/8/4 W7 R1 | VIII-A tolerant control<br>VIII-C executed recovery<br>VIII-D resource management |
| | Computation and energy limits | 4/2/2 W2 N2 | 4/0/0 W4 | 9/0/0 W7 G2 | 6/1/1 W5 N1 | 7/3/3 W4 R1 N2 | |

**Strongest setting with a named condition:** none; replay, simulation, model, or resource; hardware or recorded data; physical closed loop

Fig. 1. Degradation-source by function evidence matrix of the 244 technical publications. Rows group nine degradation sources by the navigation assumption they violate. The right column lists the response families that address each assumption, by subsection. In each cell, $a/n/m$ counts the works of the section that address the source, name it as an evaluated condition (D1 or D2), and inject or vary it against a comparator (D1). The second line splits the $a$ works into target wheeled robots (W), road vehicles (R), other ground robots (G), and non-ground, handheld, mixed, or unspecified carriers (N). Shading marks the strongest evidence setting among the $n$ works. A work appears in every source row it addresses but only in the column of its principal section. Counts describe the purposively selected sample.

### *B. The Navigation Stack and Its Degradation*

Cameras, color and depth (RGB-D) cameras, LiDAR, radar, inertial measurement units (IMUs), wheel encoders, global navigation satellite system (GNSS) receivers, and ultra-wideband (UWB) ranging supply measurements with calibration parameters and timestamps. The estimator infers robot state from these measurements and motion inputs, the mapping algorithm estimates spatial structure, the planner selects a route or trajectory, and the controller converts that reference into actuator commands. A supervisor can interrupt execution when measurements, models, or resources no longer support the current action. Deployed systems such as the Nav2 stack of the robot operating system (ROS) [38], [39] realize these functions through interacting maps, planners, controllers, and recovery behaviors, as in planner fallback [34] or behavior-tree supervision [37].

An estimator can be described through the posterior [40]

$$p(\mathbf{x}_t, \mathcal{M} \mid \mathbf{z}_{1:t}, \mathbf{u}_{1:t}), \tag{4}$$

where $\mathbf{x}_t$ is the robot state, $\mathcal{M}$ is the map, and $\mathbf{z}_{1:t}$ and $\mathbf{u}_{1:t}$ denote observation and motion-input sequences available through time $t$. Localization estimates state relative to a given reference, whereas SLAM estimates state and map jointly, commonly through visual–inertial odometry (VIO) or LiDAR–inertial odometry (LIO). Loop closures constrain revisited places, although an incorrect association can introduce correlated errors into many states. Downstream decisions may therefore require the covariance, association confidence, and age of an estimate as well as its mean.

In local planning, the dynamic window approach (DWA) searches admissible velocity commands [6], timed-elastic-band (TEB) optimization adjusts a sequence of poses and timing variables [7], and model predictive control (MPC) repeatedly solves a finite-horizon control problem. A controller using a control barrier function (CBF) constrains inputs to keep the state within an admissible set under stated assumptions [41]–[43]. For a control-affine model $\dot{\mathbf{x}} = f(\mathbf{x}) + g(\mathbf{x})\mathbf{u}$, a common condition is

$$\nabla h(\mathbf{x})^{\mathsf{T}} \big[ f(\mathbf{x}) + g(\mathbf{x})\mathbf{u} \big] + \alpha(h(\mathbf{x})) \geq 0, \tag{5}$$

where $f$ denotes the drift dynamics, $g$ maps the control input $\mathbf{u}$ to the state derivative, $h(\mathbf{x}) \geq 0$ defines the admissible set, and $\alpha$ is a suitable increasing function with $\alpha(0) = 0$.

Here, *degradation* denotes an operating change that reduces the quality or validity of the information, models, resources, or physical capabilities required for navigation. Fig. 2 distinguishes five groups of degradation sources, which can occur together within one navigation episode and which the codebook of Section III splits into the nine sources of Fig. 1.

### *C. Accuracy, Task Performance, and Integrity*

Absolute trajectory error (ATE) measures differences between estimated and reference poses after a declared alignment, whereas relative pose error (RPE) measures motion error over a declared interval. Both are commonly summarized by the root mean square error (RMSE), whose value depends on alignment freedoms, scale treatment, synchronization, aggregation, and the dependence of the reference on the evaluated sensors. Task measures include goal success, collisions, interventions, completion time, coverage, and energy. Success weighted by path length (SPL) averages over episodes the

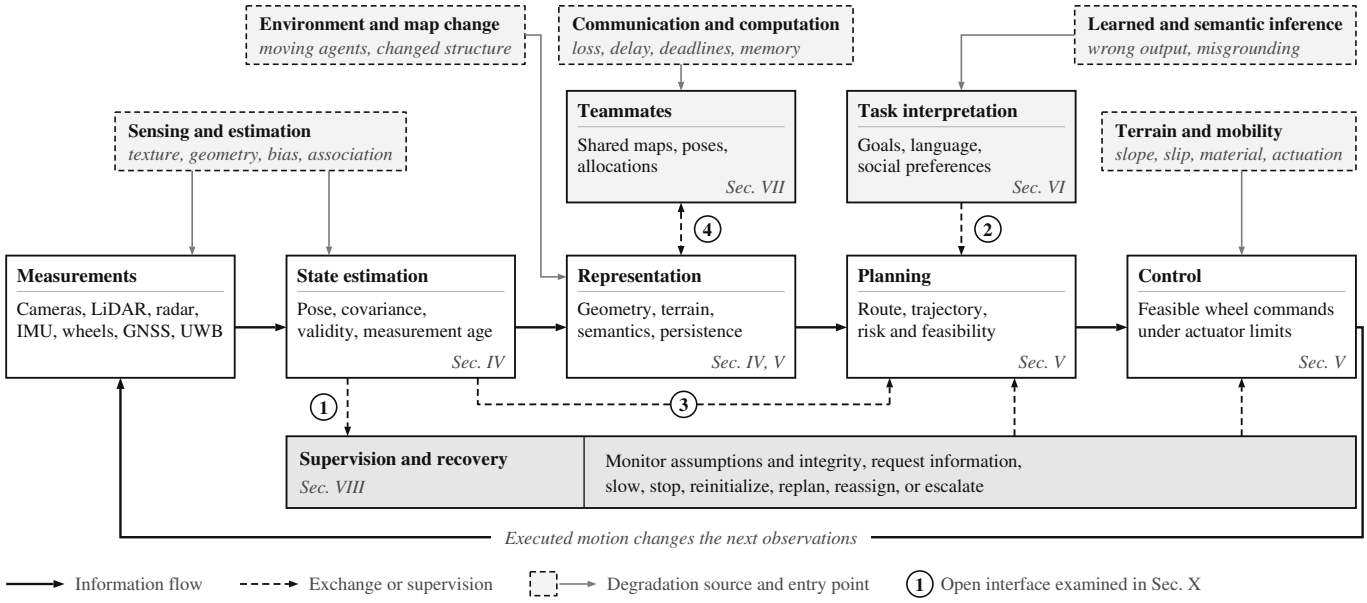



Fig. 2. Navigation functions, degradation entry points, and open interfaces. Each function box names the section that reviews it. Dashed boxes mark the five groups of degradation sources. Teammates and task interpretation supply information to the motion stack rather than bypassing vehicle constraints. The circled numerals locate the four interfaces of Section X. The dashed link at numeral 3 carries pose covariance from state estimation to planning.

shortest path length divided by the longer of the shortest and executed paths, with zero for a failed episode. Neither task success nor low ATE identifies the source of a failure or the ability to recover after an alarm.

Integrity concerns the risk of an unacceptable estimation error without an adequate warning [44]. A useful conceptual statement is

$$\Pr(E_t > A_t \ \wedge \ \text{no timely alert at } t)\,, \tag{6}$$

where $E_t$ is an error magnitude and $A_t$ is an application-specific alert limit. A protection level is an error bound associated with a specified integrity model, whereas a covariance alone is not a calibrated protection level. Particle-filter and SLAM integrity analyses account for fault hypotheses and association errors when relating alarm conditions to hazardous error [45], [46]. For navigation, the warning must also reach a response policy early enough for the motion and stopping constraints of the robot.

### D. Datasets, Simulators, and Transfer

Table III relates representative datasets and simulators to the kinds of evidence they can support.

## III. Review Method

This survey covers localization and mapping, traversability and motion control, semantic navigation, multi-robot coordination, and fault management for wheeled mobile robots operating under degraded sensing, terrain, maps, communication, or computation. It includes journal articles and main-conference papers from 2021 to 2026 that propose, evaluate, or benchmark a mechanism for these functions. Studies on road vehicles, other ground robots, or non-ground carriers are included when the inputs of their mechanism exist on a wheeled robot and its assumptions do not rest on the other embodiment. Fig. 1 and Table V count these platform groups separately.

The 245 publications were selected in purposive stages. A language-model-assisted title and abstract screening of robotics journals formed the initial sample, which theme-balanced and targeted searches extended together with works that an internal review of an earlier draft named as missing (Section S1 of the supplementary material). The corpus is dominated by IEEE publications (72%) and by works from 2025–2026 (71%). Its counts in Section IX therefore describe the selected literature rather than the field. Foundational methods from before 2021 are cited as background outside the coded corpus.

Each work is coded with a published codebook (Section S6 of the supplementary material). Degradation evidence falls into three classes: a condition injected or varied against a comparator (D1), a condition deliberately selected for its difficulty without manipulation (D2), and an incidental condition that the study reports (D3). A named degradation condition means D1 or D2. A combined attribute such as a task outcome under such a condition counts only when both parts come from the same experiment. A separate language-model session coded a stratified sample of 40 works without access to the primary codes, with a median Cohen's kappa of 0.81 over the double-coded attributes (Section S7 of the supplementary material). This value measures agreement between two language-model coding passes and is not human inter-rater reliability. Reported metrics are interpreted within the platform and protocol of each study, without pooling effect sizes across datasets.

## IV. Localization and Mapping

Localization and mapping responses to degradation differ in the information they retain and in their assumptions about geometry, contact, and correlations between estimates (Table S4 in the supplementary material).

TABLE III
REPRESENTATIVE EVALUATION RESOURCES, THEIR INTENDED USES, AND RELEVANT LIMITATIONS.

| Resource | Evaluation use | Relevant limitation |
|---|---|---|
| M2DGR [1] | Ground-robot localization across everyday sensing and motion difficulties | Estimation errors must be connected separately to task outcomes |
| M2UD [2] | Uneven-terrain localization/mapping and runtime-aware comparison | References mix RTK and offline SLAM, external checks cover selected routes, and mapping includes handheld capture |
| GEODE [47] | Degeneracy comparisons across sensor conditions | Sensor-specific coverage and severity affect comparisons, and replay does not test action-dependent recovery |
| SubT-MRS [48] | SLAM accuracy and robustness under visual, geometric, and mixed degradation, including alternating sensor loss | Sequences mix aerial, legged, and wheeled carriers, and robustness is scored offline from velocity error |
| HK-MEMS [49] | Multi-sensor localization and reference comparisons | Missing reference intervals and failed trajectories require explicit coverage reporting |
| RELLIS-3D [50] | Off-road semantic segmentation with labeled LiDAR scans and images | One site and one vehicle, with object classes rather than traversability labels |
| TartanDrive [51] | Learning off-road vehicle dynamics from multimodal driving interactions | Recorded on a joystick-driven all-terrain vehicle, without closed-loop use of the learned models |
| HM3D [52] | Simulated embodied navigation in reconstructed building interiors | Agents receive exact position and heading in static scenes without object semantics |
| Robot/network simulation [53] | Designed reception and control conditions with repeated trials | A network-in-the-loop simulation is distinct from a distributed physical deployment |

### *A. Localizability and Selective Measurement Updates*

Under geometric degeneracy, LODESTAR keeps the information that remains in the weak directions by selecting historical Schmidt–Kalman anchors from the condition of the LiDAR measurement Jacobian and choosing points by their contribution along those directions [54]. LODESTAR reports absolute pose error (APE), an ATE from the EVO tool with no stated alignment or statistic. With a common FAST-LIO2 backbone, its APE on the Long Corridor recording of the SubT-MRS dataset [48] falls from 3.113 to 1.386 m, while average desktop computation rises from 21.04 to 28.69 ms. The improvement is not uniform, since the APE on the Short exp recording is 0.429 m against 0.412 m for the baseline. Other methods characterize degeneracy explicitly, through screw-theoretic classification of translational, helical, and spherical degeneracy [55] or a DBSCAN-based detector that adapts to condition-number history and compensates with IMU motion after a startup phase [56]. Others condition it, through normal-histogram feature selection along weak tunnel directions in DR-ICP [57] or through DCReg, which preconditions the weak subspaces without changing the least-squares objective [58]. Such conditioning exploits weak but nonzero information without restoring an absent constraint or rescuing an initialization outside the convergence basin.

### *B. Complementary Sensing*

Complementary sensors differ mainly in when the second modality enters the estimate. Mixed degradation remains difficult, since no sensor-fusion entry met the success criteria in the SubT-MRS challenge, whose sequences come from aerial, legged, and wheeled robots [48]. Its LiDAR entries averaged 1.61 m ATE on mixed-degradation sequences against 0.836 m on geometric ones. Switch-SLAM adds visual constraints only when normalized Hessian eigenvalues indicate LiDAR degeneracy and falls back to inertial propagation when both modalities fail [59]. Degradation-resilient LiDAR–radar–inertial odometry fuses continuously, keeping LiDAR constraints along observable directions and using radar velocity as independent motion information in obscurant tests on a flying robot [60]. An equivariant LiDAR–radar–inertial filter estimates inertial biases and rejects radar outliers, with observability resting on an exact map [61]. Radar-centric estimators must also cope with sparse, noisy returns, as a continuous-trajectory radar–inertial estimator propagates pose uncertainty to each point timestamp and down-weights uninformative points [62], whereas Super4DR learns radar odometry by self-supervision and completes the radar map with Gaussian primitives partly guided by image depth [63]. Teammates provide another source, as RaLI-Multi corrects LiDAR–inertial drift along the detected degenerate direction with inter-vehicle UWB ranges, provided that at least one horizontal direction remains well constrained [64]. When no modality remains informative, MM-LINS hibernates the unreliable map, reinitializes, and registers the maps once they overlap again [65].

### *C. Motion Priors, Contact, and Adaptation*

Ground geometry enters estimation as a reweighting of ground and non-ground corrections in AESF-LIO, which adds no independent observation [66]. It enters as a planar prior in SE2-LIO, which parameterizes pose on $SE(2)$ as in (1) [67], or as an explicit ground-normal factor in a sparse-channel LiDAR system whose ground detector misses gradual slope changes [68]. All three designs rely on near-planar ground.

Wheel and inertial measurements bridge failures of external sensing while contact and excitation remain adequate. Wheel-aided VIO propagates inertial–wheel motion during visual failure [69], while PL-VIWO adds motion-consistency rejection and point-assisted line triangulation with its calibration held fixed [70]. Ground models extend this propagation beyond flat

floors. A quadratic motion-manifold estimator integrates wheel measurements on a local ground model that extremely uneven farmland violates [71], whereas WING learns IMU-bias and wheel-error corrections for skidding and vertical bouncing on a continuous B-spline ground manifold, yielding interoceptive-only odometry [72].

Contact uncertainty can also be estimated explicitly, as when a tire-model observer fuses wheel-level velocities by covariance intersection and excludes inconsistent wheels [30]. Time-of-flight (ToF)–wheel odometry inflates process covariance only along the remainder that the differential-drive model (2) cannot explain [27]. DURAL switches between LiDAR–inertial–UWB estimation inside a surveyed anchor region and wheel-aided estimation outside it [73]. Learned kinematic factors [32] and the hierarchical modality reweighting of Super Odometry [74] train motion priors while perception is reliable and freeze them when supervision degrades.

Lug-generated wheel traces reveal slip on suitable soil [75], whereas commanded collisions with known-map geometry constrain pose in a filter, tested on hardware with manually commanded contacts [76]. GNSS-aided IMU–odometer calibration estimates wheel scale and extrinsics but leaves vertical translation unobservable under planar motion [77]. Such calibration parameters should not absorb transient contact failure. IL-LIO learns point-to-voxel residuals and uncertainty for nonplanar geometry [78], although learned residual variance is not an independently calibrated integrity bound.

### D. Dynamic Geometry and Persistent Maps

Dynamic-scene estimators either exclude moving observations or estimate them. Exclusion keeps the static map clean but can remove much of the geometry in crowds, as in semantic rejection of person features in hospital RGB-D SLAM [79] and in GPU-free asynchronous mask propagation in NGD-SLAM, whose reported tracking time excludes concurrent detection [80]. DynoSAM instead estimates rigid-object motion in world-centric factor graphs through full-batch optimization without loop closure [81]. PG-SLAM and JPG-SLAM retain dynamic appearance through Gaussian representations, although neither runs in real time with loop closure [82], [83]. Improved rendering does not by itself establish a better obstacle map.

Across sessions, persistence enters as a survival prior on semantic clusters in ProbPer-LiLo [84] and as local and cross-session ephemerality weights for registration in ELite [85]. A lifelong LiDAR mapper instead stores positive and negative map differences, from which earlier versions can be reconstructed [86]. Chamelion learns change detection from synthesized structural changes and separates disappearance from occlusion before updating construction-site and office maps, performing best when registration error and voxel size resemble its training data [87]. For automotive radar, multi-session loop verification and a temporal occupancy grid retain persistent points for later localization under a planar-trajectory assumption [88]. NVMS-SLAM keeps normal-bearing surfaces apart to avoid merging opposite faces of thin walls and refines alignment offline [89], while NeuSE updates object maps after layout changes within the categories on which it was trained [90].

### E. Prior-Map and Cross-Modal Localization

Localization against a compact or structural prior reduces the online representation but becomes sensitive to changes in the mapped environment. ERPoT matches sparse scans to polygon maps after ground removal [91], Area Graph localization matches filtered scans to walls and doors after WiFi and barometric initialization [92], and building information model (BIM) localization estimates as-designed discrepancies before weighting structural constraints [93]. Orchard trunk maps suppress seasonal foliage at the price of making global recovery depend on adequate trunk support [94]. An uncertainty-aware outdoor localizer weights normal distributions transform (NDT) matching by estimated covariance and gates GNSS, a rule that can reject accurate fixes after extended LiDAR loss [95].

Relocalization obtains pose hypotheses from appearance, through cross-attention image retrieval on pushed-robot recordings [96] or distilled LiDAR–camera place recognition [97]. Structural hypotheses come from object-level scene graphs that tolerate staged object moves [98], rendered point-cloud, mesh, or neural-radiance maps [99], and density maps for loop closure that can inherit front-end degeneracy [100]. Retrieval provides a candidate rather than a verified pose.

Cross-modal matching relaxes the requirement that map and query share a sensor. CMRNext casts camera-to-LiDAR matching as optical flow followed by perspective-n-point estimation and transfers across camera setups, although it is trained on driving data, does not yet run in real time, and receives an initial pose for every frame [101]. OMCL scores Monte Carlo hypotheses with open-vocabulary features ray-cast from a three-dimensional map and can initialize from a language description, provided that odometry is map-scaled and the map features still match the scene, since the map is not updated online [102]. LiDAR Teach, Radar Repeat localizes forward-looking 4D radar against LiDAR keyframes from a teaching pass, with registration errors driving label-free fine-tuning [103]. Its route repetition on a four-wheel differential-drive platform covers smoke, traffic, and structural change over months, with obstacle avoidance left to other sensors.

### F. Cooperative Estimates, Uncertain Maps, and Evaluation Resources

Cooperative estimators add constraints while introducing correlations. CoVOR-SLAM corrects monocular scale with inter-agent ranges in a similarity-pose graph [104], while a Lie-group estimator tracks the cross-covariances among wheel, inertial, and marker estimates over near-straight indoor motion [105]. For uncertainty in the map itself, semantic neural processes predict elevation with an uncertainty that still requires calibration [106], whereas E2-BKI propagates evidential semantic uncertainty through geometry-aligned kernels [107]. A multi-fidelity Gaussian process (GP) explorer on an aquatic robot weights field samples by localization uncertainty [108].

Robot-mounted lighting in LUCID improves visual appearance without necessarily improving odometry [109]. AirSLAM combines learned point and line features with map reuse and reaches 40 Hz on an embedded GPU under illumination change, although it reduces to a point-feature system where structural lines are absent [110].

### G. Synthesis

The evidence for these responses stops at the estimate, since only 1 of the 67 studies reviewed in this section reports a task outcome (Table V), while most also evaluate on carriers other than a target wheeled robot.

## V. Traversability, Motion Planning, and Control

Under degradation, the feasibility of planned motion depends on terrain support, obstacle motion, state uncertainty, actuator limits, and the age of control information (Table S5 in the supplementary material).

### A. Terrain Support and Traversability Costs

Learned costs describe where the robot can go. CADNet predicts heading-dependent accessible range from LiDAR history for a TEB planner on a SCOUT robot [111], whereas GA-Nav projects attention-based terrain segmentation into a cost map combined with elevation [112]. A risk-aware learner labels traversed footprints and weak obstacles before predicting unvisited cells, with predictions evaluated only on manually driven recordings [113]. Such learners depend on off-road data such as the TartanDrive dynamics recordings [51] and the RELLIS-3D campus scans, on which SalsaNext segmentation reaches 40.20% mean intersection over union against 59.5% on an urban benchmark [50]. Physical models describe whether the ground can carry the robot. PE-RRT* infers hidden support planes in vegetation with multivariate Gaussian processes and tracks the route with nonlinear MPC [114], while a multilevel planner checks settling, support polygon, and body collision for a tracked robot, with its wheeled variant tested only in simulation [115]. An anisotropic explorer penalizes motion by slope and travel direction with a quasi-static Euler Cost that omits slip and suspension transients [116]. A footprint label, a support plane, a stability margin, and a cross-slope penalty are not interchangeable estimates of traversal risk.

Recent planners condition terrain risk on approach direction and uncertainty and embed it in optimization. SPARTA predicts angle-conditioned risk distributions as reusable Fourier functions from a fully observed prior point cloud [117]. A visual-foundation-model approach learns cost, speed, and novelty maps from expert driving for model predictive path integral (MPPI) control, with field trials in which operator corrections expose residual failures that offline trajectory agreement conceals [118]. STEP combines risks such as collision, step, tip-over, and slip on a 2.5D elevation map into a conditional value at risk (CVaR) map for path search and MPC, with localization error entering through the elevation-map variance [119]. Its onboard runs on a Husky and a Spot quadruped report explored extent without failure rates. EVORA plans with MPPI against the CVaR of learned traction distributions and penalizes out-of-distribution terrain, succeeding in all five indoor trials at its best risk level on an RC car whose pose comes from motion capture [120]. A terrain-adaptive $\epsilon$-constraint MPC minimizes path length while bounding a pose-stability cost adjusted online from terrain descriptors, with a sparse GP correcting its car-like dynamics [121]. Its success falls from 94% in simulation to 80% outdoors on a Scout Mini, although it still exceeds the 67% of MPPI and 73% of GAKD in the same outdoor comparison. TRAIL learns an implicit field of terrain attributes from RGB and LiDAR and optimizes path geometry and speed along its gradients [122].

### B. Exploration with Information and Geometry

Exploration chooses where to improve the map while preserving the ability to localize, as FIT-SLAM 2 does by attaching frontiers to SLAM keyframes, adapting a traversability roadmap after pose changes, and rejecting edges with low expected landmark information [123]. With a common controller and simulator pose, completion takes 880 s versus 1118 s for FAEL in Cave but 1809 s versus 1706 s in Corridor. Fisher-information evaluation also raises mean planning time from 20.8 to 63.0 ms. Among planners that score uncertainty directly, confidence-rich exploration combines particle-trajectory entropy with occupancy confidence and executes the loop on hardware [124], whereas an uncertainty-frontier planner scores coverage with signed relative entropy under known orientation [125]. A further group changes the representation of unexplored space, through safety levels for unknown regions in LRAE [126] and octree category distributions for semantic exploration [127]. FRTree instead builds overlapping free polytopes and checks that each region can contain the robot geometry, with tests on a quadruped rather than under wheeled curvature limits [128]. A SLAM-aware planner inserts informative loop closures over a supplied topometric graph [129]. Potential-field MPC integrates frontier and unknown-space costs on a six-wheel skid-steer robot, although without systematic baselines [130].

### C. Learning, Prediction, and Interaction

Learning enters planning at different points. NAMR-RRT learns where a risk-aware tree samples [131], whereas prescient navigation passes multimodal pedestrian hypotheses to nonlinear MPC, which may exhaust its time budget [9]. A learned relaxation of optimal reciprocal collision avoidance (ORCA) predicts escape direction and avoidance responsibility [132]. SCOPE predicts future occupancy-grid distributions from robot motion, tracked objects, and a local static map and inserts their mean and uncertainty into the costmaps of existing planners, with distillation enabling embedded inference [133]. However, its navigation comparisons are mainly simulated, while its constant-velocity robot model fails during abrupt rotation. Homotopy-driven MPC optimizes distinct trajectory classes in parallel from externally tracked poses [134], a form of candidate diversity that reduces local inconsistency

without compensating for a missed pedestrian or an incorrect interaction model.

Learned local policies encode observation history implicitly. Visual proximal policy optimization (PPO) adds a motion-perceptibility auxiliary task [29], depth-based recurrent PPO is compared with DWA and TEB in simulation [8], and a LiDAR actor–critic controller uses spatiotemporal attention under adaptive Monte Carlo localization (AMCL) on a prebuilt map [135]. HEIGHT separates pedestrians from static obstacles in a heterogeneous spatio-temporal graph and transfers from a low-fidelity simulator to four everyday environments, although its deployed obstacle input assumes a map and reliable localization [136]. A rough-terrain sim-to-real policy trained with terrain scaling and pose noise still fails under vibration and inaccurate reconstructed geometry [137]. Lifelong navigation, in turn, learns only where an initial DWA policy supplies useful experience [138]. As the learned-navigation review notes [17], the training data and observation representation determine which disturbances such controllers can accommodate.

### D. Safety Filters and Explicit Risk Conditions

A LiDAR-based CBF safety filter fits a GP to current obstacle edges and uses its mean and gradient in the condition (5) of a quadratic program, with an invariance argument that excludes the GP variance [43]. Other barrier designs add structure to the motion or the learning, through circulation with hold logic [139], CBF-corrected movement primitives [140], and high-order barrier penalties learned alongside soft actor–critic [141]. Others address the conditions under which the barrier must hold, through high-order filtering of iterative linear quadratic regulator (LQR) commands under input bounds [42] and disturbance estimation on a tracked robot [142]. Their evidence ranges from external pose and offboard computation to numerical studies.

Other controllers constrain an explicit risk event, through noisy barrier samples in a distributionally robust CBF [143], selectable CVaR levels in crowds [144], or a risk-aware zone expanded under input saturation, whose score is not a calibrated probability [145]. Safe Horizon MPC samples obstacle trajectories, applies a plan only after slack and support checks, and otherwise decelerates [146]. In 100 Gaussian eight-pedestrian simulations with a joint-risk specification of 0.05, its mean traversal takes 16.0 s (standard deviation 1.9 s) versus 18.4 s (2.4 s) for chance-constrained MPC, with maximum estimated plan collision probabilities of 0.0092 and 0.0019. Calibration offers a third route, in which Perceive With Confidence calibrates obstacle inflation over sampled environments for a planner with an inevitable-collision-set constraint on a legged robot [147], whereas conformal crowd navigation limits cumulative intrusion into calibrated pedestrian regions on a Mecanum robot [36]. MonoMPC learns the distribution of minimum clearance for candidate controls from monocular depth, without temporal memory, and minimizes a distribution-based risk [148]. In Jackal trials on the same estimated depth, its collision rate is 6.6% against 25% for a navigation stack at a similar mean speed and 48.3% for a faster setting.

### E. Tracking, Mission Costs, and Network Support

A GP-MPC controller learns residual dynamics offline from field data and uses only the GP mean [149], while predictive learning refines repetitive Mecanum trajectories under external UWB pose [35]. Communication loss calls for a different response. Delay-aware MPC predicts current states from delayed packets and buffers future controls [150], whereas vehicle–cloud MPC runs onboard and remote optimizers in parallel and stops on timeout [151]. Mission-dependent costs add another reason to change a route, as in infrastructure-based hazard inference added to a cost map [152] and in payload- or sensing-aware MPPI on a compressed scouting map [153].

Several controllers offer guarantees or savings under restricted models, including prescribed-time spatiotemporal tubes [154] and memory-event-triggered MPC, which reduces repeated solves for the no-slip Ackermann model (3) in hardware-in-the-loop tests [155]. Curvature-constrained vector fields co-design motion and saturation on an Ackermann vehicle with motion capture [33], while universal constrained tracking filters references that exceed prescribed state bounds [156].

### F. Synthesis

Support and stability models describe physical failure modes such as sinkage, tipping, or slip, which learned terrain costs capture only through appearance and traversal experience. Predictive constraints with a stated risk event instead make admissibility around moving obstacles testable under their state, dynamics, and update assumptions. Unlike the estimation studies of Section IV, these studies report task outcomes under a named degradation condition in the same experiment more often than the studies of any other function (Table V).

## VI. Semantic and Language-Guided Navigation

The output of a semantic navigation system, whether a room, an object instance, frontier scores, a path, a cost layer, or an action sequence, determines how a reasoning error propagates into planning and control (Table S6 in the supplementary material).

### A. Semantic Memory and Scene Structure

Persistent semantic maps support relational queries but must be updated when objects move. HOV-SG organizes a mapped building into floors, rooms, and objects for hierarchical retrieval [157], while DualMap searches stable anchors globally before checking objects locally [158]. ConceptGraphs builds an open-vocabulary 3D object graph from 2D foundation models, whose node captions a language model reads to select targets [159]. On a Jackal, the language model redirects search to a plausible container when the target has moved, in demonstrations without success rates. OpenIN updates the existence confidence of a carrier–object pair when an expected object is not observed [160]. Such updates must distinguish removal from occlusion or poor depth. osmAG-LLM instead assumes from the outset that objects move,

keeping a lightweight text-based area graph for context, asking a language model for likely locations, and confirming targets by online detection [161]. In 30 building trials it succeeds end to end in 21 and reaches a correct position in 27, with detection errors causing most failures.

Goal-conditioned maps store less reusable structure, as VLFM projects image–language similarity into a frontier value map until a detector supplies the goal [162]. On simulated Habitat-Matterport 3D (HM3D) object search, where success means stopping within 1 m of a target instance in at most 500 steps, VLFM reaches 52.5% success and 30.4% SPL. VLN-Game builds an object-centric three-dimensional map and uses a game-theoretic vision-language model (VLM) to disambiguate attribute-rich descriptions [163] in HM3D scenes, whose own point-goal benchmark gives simulated agents exact position and heading [52]. SysNav reserves VLM calls for room-level decisions and leaves viewpoint selection and waypoint execution to geometric modules [164]. Multimodal spatial language maps add image and audio features to an offline map, although a remembered sound marks where it was heard rather than its source [165]. Temporal scene-object graphs report corner trapping from missing depth and episodes that end with the target out of view among their physical failures [166].

### *B. Geographic Grounding, Priors, and Learned Actions*

Outdoor instructions often refer to places beyond the sensor horizon. OPEN associates visual semantics with OpenStreetMap and replans locally when road geometry differs from the map [167]. G-DRAGON constrains generated identifiers to existing map entities before composing a behavior tree [168], whereas CAUSALNAV retrieves destinations from an object–building–cluster graph and combines global routing with constrained local control [169]. HAM-Nav associates live landmarks with an imperfect hand-drawn map and predicts omitted ones before A*/TEB execution [170], whereas Mobility VLA retrieves a goal frame from a demonstration tour and routes over an offline graph without exploring [171].

Navigation foundation models reuse pretrained components or pooled robot data. LM-Nav parses an instruction into landmarks with GPT-3, grounds them with CLIP in a graph built by manually driving a Jackal, and finds a successful walk for 85% of 20 instructions over more than 6 km with one disengagement [172]. GNM trains one image-goal policy on data pooled from six robots and controls four robots zero-shot, including a quadrotor, still reaching goals when its angular velocity is clipped or its camera is displaced [173]. ViNT scales the data to over 100 h from eight platforms and drives an unseen Go 1 quadruped at a higher inference cost than feedforward convolutional policies [174]. NoMaD unifies goal-free exploration and image-goal navigation in one diffusion policy on a LoCoBot with 15 times fewer parameters than the strongest exploration baseline [175]. Learned policies map language and images directly to motion, as in LeLaN, which trains a last-mile policy from labeled robot and action-free video [176]. In 150 trials with 28 initially visible objects, its success within 0.2 m is 0.89 versus 0.67 for OWL-v2 with ZoeDepth. In a separate ablation, collision training raises arrival with obstacles from 0.13 to 0.60 in 15 trials but lowers success without obstacles from 0.89 to 0.77. Uni-NaVid compresses multiscale video history before predicting discrete primitives whose vocabulary bounds the available actions [177], whereas LGX prompts a language model for exploration directions and falls back to random exploration after an invalid response [178].

### *C. Candidate Selection and Behavioral Constraints*

VL-TGS generates LiDAR-based trajectories, projects them into an image for GPT-4V ranking, and follows the choice with DWA while later decisions are pending [179]. However, its ranking cannot recover a route that the generator never proposes. Sem-NaVAE applies the same generate-and-select structure without a map, scoring conditional variational autoencoder (VAE) proposals with lightweight open-vocabulary segmentation and reporting 90% success on 120–240 m campus and park routes with few repetitions per route [180]. VAMOS adds an embodiment filter, in which a per-robot affordance model trained in simulation re-ranks the image-space paths proposed by a VLM planner [181]. With the planner unchanged, the affordance model of a wheeled robot raises its success on a ramp-or-stairs task from 6 to 9 of 10 runs.

Social and behavioral constraints separate preference from collision geometry at different levels of the stack. Cue-triggered judgments become social costs for DWA in VLM-Social-Nav [182], while NORM-Nav writes instruction-parsed semantic, directional, and speed layers into standard costmaps [183]. VLM-informed path selection ranks clustered detours before an ORCA controller computes velocities [184], whereas GSON supplies group labels to a mid-level planner above a barrier-constrained predictive controller [185]. OLiVia-Nav distills large-model social reasoning into lightweight encoders that are updated online, with evaluation against expert trajectories and personal-space violation rather than collision rates [186]. Safe-VLN masks predicted waypoints with simulated occupancy and selects a new waypoint after a collision occurs [187]. SocialNav-SUB shows that strong VLMs still trail human consensus and position-based rules on social scene questions [188], while VLM-GroNav revises terrain costs and waypoints from visual and proprioceptive exemplars [189]. A grouping or grounding error can thus alter the route even when the local controller avoids detected people, whereas an occluded pedestrian can invalidate the controller itself.

### *D. Memory, Timing, and Embodied Execution*

Semantic decisions remain valid for different durations. TARIC triangulates intermittent goal cues into a particle memory that keeps an executable goal direction during cue-free travel, although the memory drifts over long cue-free stretches [190]. Other systems align the timing of reasoning with the scene, as SysNav queries at room transitions, GSON uses updated keyframes, and the social path selector associates asynchronous responses with current candidates [164], [184],

[185]. VAMOS plans at 1 Hz on a laptop GPU [181], a rate that a fast local controller can bridge only while it retains a valid action. Embodiment then determines how a semantic choice becomes motion. VLFM uses different low-level controllers in simulation and on Spot [162], Uni-NaVid runs remote inference on a quadruped [177], and LeLaN executes on a wheeled base [176]. Table II lists the recovery motions that each wheeled drive allows.

### *E. Synthesis*

Candidate selection confines semantic errors more directly than cost layers, which keep geometry in charge of the final motion but can still misdirect the route, whereas direct policies depend on their history and action vocabulary. Semantic correctness and collision avoidance therefore require separate evaluation. As in Section V, most studies report task outcomes, while fewer than half report one under a named degradation condition in the same experiment (Table V).

## VII. Multi-Robot Coordination

Since robot teams exchange frontiers, maps, observations, poses, or position histories, communication degradation turns local operation and shared-state consistency into separate objectives (Table S7 in the supplementary material).

### *A. Task Allocation and Spatial Responsibility*

Allocation reduces redundant work through a common state. Centralized planners partition merged maps into dynamic Voronoi cells in CURE [191], assign known targets over a sparse Delaunay graph with priority yielding in DMT-CPP [192], or fix a robust task schedule before the mission under uncertain agent capabilities [193]. Distributed methods reduce negotiation without removing shared state, as CQLite exchanges selected Q-values and frontiers [194], whereas LPFE predicts teammate frontier choices from a common map and last known positions [195] and 4CNet predicts map structure from shared trajectories [196]. Such inference avoids messages only while robots hold compatible inputs.

Region-level allocation makes the shared state coarser, as URGE abstracts the map into subregions with exploration states and allocates them by jointly minimizing path cost and balancing load [197]. Its field runs with three robots use an ad hoc mesh network in a forest, a parking lot, and a tunnel. Terrain-aware partitioning (TAPM-IVP) extends spatial responsibility to uneven ground with communication-radius-constrained regions and traversability-aware subgoals [198]. In ten simulated trials of Scene 1, each robot succeeds in 90% of trials, compared with 40% and 30% for an ablation that removes both partitioning and greedy subgoal selection.

### *B. Compact Sharing and Distributed Estimation*

Compact representations trade bandwidth against what teammates can infer. MR-TopoMap shares topological vertices with place descriptors and keeps occupancy local for avoidance [199]. Cross-rank exchanges position histories and assigns directions at intersections [200], while WiSER-X infers relative positions from radio pings, which require an interpretation when they are missing [201]. Map-sharing systems decide what to transmit and when to fuse it, as SlideSLAM buffers exchanged object observations until a loop closure fixes the frame transform [202], whereas Commerge minimizes the scans transmitted for central submap registration without adapting to radio capacity [203]. Feedback-guided hierarchical transmission sends a pose-and-factor skeleton, prioritized geometry, and semantic labels in layers and reallocates bandwidth each round from back-end signals, although its authors note that round-wise guidance can become stale under asynchronous links [204]. PROMO-Orchard gates submap merging with orchard priors and posterior checks [205].

Distributed estimators must also handle delayed and repeated information. iMESA keeps local incremental factor graphs and updates consensus variables through a two-stage exchange that must complete once started [206]. riMESA, by the same authors, removes this blocking requirement with edge-based consensus under the alternating direction method of multipliers (ADMM), robust local outlier rejection, and discarding of failed exchanges [207]. Multi-Proxy combines lightweight descriptors for progressive loop detection with asynchronous ADMM pose-graph optimization and continues under simulated packet loss and latency, where a two-stage Gauss–Seidel baseline fails to converge, provided that front-end odometry stays stable [208]. Parallel graph inference separates current-state from historical-message inference [209]. Inter-robot loop closures also require outlier rejection. Kimera-Multi rejects outlier loop closures with distributed graduated non-convexity (GNC) on recorded Jackal data, with robots kept constantly in communication range [210]. LAMP 2.0 instead queues pose graphs and keyed scans while a robot is out of range and sends them as a batch on reconnection [211]. Its base station filters loop closures by consistency checks or GNC and keeps map errors below 4 m on replayed data from Husky and Spot robots. Swarm-SLAM runs decentralized inference with sparse loop prioritization online on a heterogeneous physical team [212], whereas foundation-model loop closing anchors monocular scale with metric odometry, evaluated by replay on a Jetson [213]. Radio geometry adds constraints with their own assumptions, as in cooperative range-only SLAM [214] and in formation-controlled UWB anchors whose geometry competes with obstacle clearance [215]. A prior-graph extension of single-robot SLAM-aware planning [129] adds informative loop edges to multi-robot coverage paths, although its approximation bound concerns the graph objective rather than physical loop correspondences [216].

### *C. Rendezvous, Disconnection, and Reconnection*

Communication can become part of the motion objective. A rendezvous-aware explorer creates backtracking frontiers from decaying trajectory records [217], whereas a communication-constrained controller schedules sub-team meetings that require full-team agreement to update [218]. CoCoPlan treats team-wise communication events as decision variables and co-optimizes them with task assignment through branch-and-bound search [219]. In simulations with up to 100 robots, it

reports a 22.4% higher task-completion rate and 58.6% lower communication overhead than the compared protocols. MEF-Explore derives rendezvous from entropy fields and assumes that low-rate position exchange is always available, exploring 21.32% faster with a 16.67% higher success rate than the baseline on three TurtleBot2 robots [220].

During disconnection, robots can continue locally and reconcile later. ROAM continues planning from local maps during observed Wi-Fi cutoffs at the cost of revisits [221]. In the DARPA Subterranean Challenge, Team CSIRO Data61 had every tracked, legged, and aerial robot solve the same multi-agent pose graph from data exchanged by store-and-forward messaging, with agents returning to base when mission-critical data remained unsynchronized [222]. Team Explorer's wheeled robots and drones broadcast a ledger of submaps and fetch missing ones from whichever robot holds them [223]. CERBERUS instead merged odometry and sensor data from ANYmal quadrupeds and aerial scouts at a base station while they were in range, with robots backtracking to connectivity or returning after a time budget [224]. These systems report scored competition runs rather than repeated trials. Correlation-aware collaboration, evaluated on recorded road-vehicle data, exchanges states, covariances, and cross-covariances of common landmarks before Kullback–Leibler averaging and Schmidt–Kalman updates [225]. Failure-aware target tracking fuses target estimates by covariance intersection after reconnection and lets a permanently isolated aerial tracker signal a hazard [226]. The communication-aware CBF controller relaxes conflicting constraints and drops unreliable links, with recovery evaluated under partial connectivity [53]. Planned synchronization makes observation an action, scheduling check-ins in which a TurtleBot pair stops and pans to observe each other [227]. A range-based integrity monitor reconstructs shared localization errors with distributed ADMM [228].

### D. Motion Interaction and Persistent Service

Coordination also occurs below the allocation layer, where RL-RVO learns a distributed velocity policy from reciprocal velocity-obstacle features with external localization [229], while predefined-time formation control compensates modeled actuator faults over a fixed graph [230]. Compensation-based MPC estimates model discrepancy online for four motion-captured nonholonomic robots and leaves deadlock to future work [231]. Reachability-based platoon control preserves separation under bounded inputs and assumes reliable communication [232], whereas distributed barrier control adds a coordinated deadlock response on physical vehicles [233]. In mEclares, evaluated with quadrotor agents and a ground charging rover, a scheduler reserves return and landing opportunities from energy and timing conditions, with agents falling back to committed paths after disconnection [234].

### E. Synthesis

When links fail unpredictably, asynchronous optimization and local planning sustain activity without a meeting schedule. When connectivity can be planned, rendezvous and communication-aware allocation trade travel for timely exchange. Fewer than a quarter of the studies reviewed here report a task outcome under a named degradation condition in the same experiment, 2 of them on physical robots (Table V).

## VIII. Fault Tolerance, Recovery, and Long-Term Deployment

Fault detection, isolation, accommodation, and restoration need not occur together, since a controller may tolerate bias of unknown source (Table S8 in the supplementary material).

### A. Fault Estimation and Tolerant Control

Analytic redundancy provides diagnosis without duplicating every sensor. Kinematics-based detection compares position, angular velocity, and acceleration through transport relations, with faults injected into recorded rover data [235]. A linear parameter-varying (LPV) vehicle diagnoser combines zonotopic set-membership estimation with invariant-set bounds, in which separation from the healthy set establishes detectability and overlap between fault classes prevents unique isolation [236]. Tolerant controllers instead accommodate faults in specified channels, each relying on a different remaining capability. Mixed-gain prescribed-performance control tolerates partial wheel-effectiveness loss [237], whereas virtual-error control tolerates heading and wheel-speed faults but assumes exact position [238]. Adaptive sliding-mode control handles injected actuator faults and saturation on industrial hardware as long as some actuation remains [239], while fixed-time adaptive tracking accommodates dead zones and modeled actuator faults in MATLAB studies [240]. An observer-based event-triggered controller reconstructs modeled faults before issuing updates in hardware-in-the-loop tests of a low-speed, no-slip vehicle [241].

### B. Integrity and Failure Monitoring

Monitors differ in the failure event they characterize. Reliable Monte Carlo localization (MCL) estimates a latent localization-success state alongside pose and triggers free-space global relocalization [242]. Integrity analysis for extended Kalman filter (EKF) SLAM accumulates landmark fault risk into the integrity risk (6) [46]. In its recorded example, this risk crosses the threshold only after localization fails. A prior-line-map protection level predicts estimation error, with operational alerting left for future work [243]. Monitors of learned policies use statistical alarm definitions, as conformal anomaly detection controls the nominal false-alarm rate under exchangeability [244], while Sentinel pairs fast action-consistency monitoring with slower vision-language progress assessment [245]. PAAD predicts anomalies along a planned path on field recordings in which people reposition the robot after failures [246]. RADIUM moves failure prediction before deployment by sampling environment parameters from a distribution conditioned on failure and repairs the policy on the discovered cases, reporting up to five times higher robustness for a repaired vision-in-the-loop policy on a 1/10-scale car [247].

### C. From Diagnosis to an Executed Response

The response must match the location of the fault in the stack. In the estimator, fault-tolerant prior-map collaboration excludes faulty observations, inflates faulty-landmark covariance, and specifies reinitialization for odometry faults without evaluating an executed reset [225]. At the planning level, a GP predictor of safe-corridor optimizer failure triggers a stop–reposition–resume sequence [248]. With the same planner in simulation, recovery yields eight successes in ten runs against three in ten in a dense world with 1 m doorways, whereas both variants succeed in all ten runs in a sparse world with 2 m doorways. However, the predictor posterior concerns planner failure rather than collision. At the level of the controller and supervisor, a fault-aware CBF controller optimizes over candidate sensor or actuator fault sets in numerical examples [249], whereas FORTRESS caches slow semantic reasoning for runtime reach-avoid fallback on a quadrotor [250].

Implemented recovery policies differ in what they restore, from localization to task feasibility. A nonlinear MPC (NMPC) visual navigation system stops after tracking loss and attempts relocalization [31], whereas behavior-tree supervision on CENTAURO chooses among contact repositioning, reversing, replanning, and human assistance [37]. Task-level policies change the mission instead, as resilient temporal-logic planning relaxes task windows or removes tasks when local correction fails [251] and predictive swarm fault tolerance withdraws an affected robot to maintenance before complete failure [252]. Local recovery acts on the trajectory or the observations, through hybrid-A* reinitialization of an infeasible local trajectory in RTEB [34], return toward familiar visual states in SCALE [253], and repulsive corrections from estimated depth to a visual policy in CARE [254]. However, neither familiarity nor estimated depth supplies a clearance certificate. Help Me Through learns viewpoint actions from about 50 h of human demonstrations, or 20 h followed by online fine-tuning, to keep monocular SLAM tracking through texture-poor transitions, with a physical test that runs the policy on live images while a person teleoperates path planning [28]. When a robot cannot restore a capability, the team can reassign its task, as in energy-aware allocation under staged capability loss on Robotarium [255] and in R-FAC, whose log-sum-exp value factorization keeps search policies effective when members fail without notification, assuming perfect target detection [256].

### D. Resources and Sustained Operation

Resource choices determine whether a response remains available. CODEI selects bodies, sensors, algorithms, and computing units from catalogs by coupling simulated planner queries to characterized perception errors [257]. Joint runtime control of motor speed and processor voltage and frequency on a wheeled rover reduces total energy by up to 36.34% relative to prior methods, with measurements showing that computation can dominate the energy of slow motion [258]. Fleet availability adds charging constraints, as in cyclic scheduling of operating and charging periods under a station budget [259] and in persistent-monitoring routes that minimize team size under revisit-latency and recharging limits with deterministic discharge [260]. A ten-home assistive deployment combines docking rules, cloud orchestration, and remote supervision, with internet problems and localization failures among the causes of interruption [3]. A cloud–edge system offloads mapping and trajectory generation over 5G while keeping MPC tracking and collision avoidance on Jetson-equipped Ackermann robots, falling back to a leader-based local network [261].

Maintenance and learning form a second operational loop. Predictive data acquisition selects new training observations for teach-repeat-learn navigation from visual matching ambiguity [262]. SELFI stabilizes deployment-time reinforcement learning with a model-based objective and counts interventions during learning [263], while LiReN combines foundation-model adaptation with pseudo-resets, keepout rules, charging, and help escalation [264]. Counterfactual replay selects among adapting policies at the cost of between-trial computation [265].

### E. Synthesis

Recovery quality in long-term service concerns both restored operation and the interruptions needed to achieve it, which depend on charging, energy, network access, offboard computation, and human assistance. The studies reviewed here contain 9 of the 14 integrity or failure-risk assessments in the sample (Table V), mostly with injected or simulated faults.

## IX. Evaluation Practice

Since most evidence gaps in Table IV concern how a result was evaluated rather than whether a mechanism exists, this section compares evaluation practice across functions.

Evaluation of autonomy under degradation must relate component outputs to executed behavior. The comparison below follows three axes, namely the level at which performance is measured, the conditions under which degradation is tested, and the event and assumptions behind a risk or diagnostic claim. Fig. 3 connects the three axes through one time-aligned navigation episode with a degraded interval.

### A. Measurement, Test Conditions, and Claim Interpretation

*Component measures* include trajectory, mapping, segmentation, prediction, and tracking error together with runtime and classification scores, whereas *task outcomes* concern completion, collision, coverage, traversal time, intervention, and service availability. *Degradation evidence* describes an experimental condition rather than a metric, in the classes D1–D3 of Section III. A matched nominal control is a stricter variant of D1, used below as a sensitivity definition. For *risk and diagnostic assessment*, the defined event matters, since classifier recall, nominal false-positive control, protection-level coverage, and undetected hazardous error are not interchangeable, as reliable MCL, conformal anomaly detection, and EKF-SLAM integrity illustrate [46], [242], [244].

TABLE IV
RESPONSES TO DEGRADATION BY NAVIGATION FUNCTION, THE CONDITIONS UNDER WHICH THEY HELP, AND THE MAIN EVIDENCE GAPS, WITH REPRESENTATIVE STUDIES.

| Function | Main responses | Conditions for benefit | Main evidence gaps |
|---|---|---|---|
| Localization and mapping (Sec. IV) | Selective updates, complementary sensing, motion and contact priors, relocalization, map revision [27], [54], [59] | Surviving observable geometry, independent measurements, credible supervision for learned priors, later overlap [58], [65], [74] | Mostly recorded or replayed data. Degeneracy and similarity scores differ in meaning. Revised maps rarely drive executed responses [48], [87] |
| Traversability, planning, and control (Sec. V) | Terrain costs, information-aware exploration, learned prediction, safety filters and risk constraints, delay-aware tracking [43], [119], [123] | Accurate pose and terrain coverage, priors and loop opportunities, training coverage, known state and admissible inputs [117], [129], [137] | Estimated rather than measured collision risk. Few simulation-to-hardware comparisons. Pose uncertainty rarely propagated into terrain decisions, its effect untested [120], [146] |
| Semantic navigation (Sec. VI) | Semantic memory, geographic grounding, learned actions, language-derived cost layers, candidate selection with embodiment filters [160], [167], [181] | Valid map updates, coverage of prior maps or tours, feasible proposed candidates, a local controller valid during pending inference [171], [179] | Incorrect or stale reasoning rarely injected. GPS, cloud, and map-preparation dependencies. Few physical trials per route [180], [188], [189] |
| Multi-robot coordination (Sec. VII) | Allocation and spatial partition, compact sharing, distributed and asynchronous estimation, rendezvous, estimate repair after reconnection [197], [204], [219] | Compatible shared inputs, reachable encounters, stable front-end odometry, tracked correlations [195], [208], [225] | Mostly ideal or modeled communication. Acceptance of stale or biased content after a partition untested on moving teams [207], [211] |
| Fault tolerance and recovery (Sec. VIII) | Analytic redundancy and tolerant control, integrity and runtime monitoring, executed recovery, resource and service management [46], [235], [248], [258] | A modeled fault channel with remaining actuation, a monitored event that matches the response, trustworthy state for the recovery maneuver [239], [242] | Injected or simulated faults. Restarted or censored trials. Interruptions and human assistance seldom reported [31], [37], [249] |

TABLE V
CODED EVIDENCE ATTRIBUTES OF THE 244 TECHNICAL PUBLICATIONS, BY THE SECTION THAT REVIEWS EACH PUBLICATION. ATTRIBUTES OVERLAP, AND THE CONTEXT REVIEW IS EXCLUDED.

| Attribute | IV | V | VI | VII | VIII | All |
|---|---|---|---|---|---|---|
| Publications | 67 | 56 | 36 | 45 | 40 | 244 |
| *Measurement level* | | | | | | |
| Component measures | 67 | 32 | 16 | 32 | 23 | 170 (70%) |
| Task outcome | 1 | 44 | 33 | 25 | 20 | 123 (50%) |
| Physical task outcome | 1 | 25 | 24 | 15 | 9 | 74 (30%) |
| *Degradation evidence* | | | | | | |
| Named condition (D1 or D2) | 61 | 46 | 19 | 27 | 34 | 187 (77%) |
| Injected or varied, D1 | 16 | 15 | 6 | 13 | 11 | 61 (25%) |
| Selected only, D2 | 45 | 31 | 13 | 14 | 23 | 126 (52%) |
| Incidental only, D3 | 1 | 1 | 2 | 4 | 2 | 10 (4%) |
| *Same experiment* | | | | | | |
| Named condition and task outcome | 1 | 36 | 17 | 11 | 16 | 81 (33%) |
| D1 condition and task outcome | 0 | 12 | 6 | 5 | 4 | 27 (11%) |
| On a physical robot | 1 | 20 | 13 | 2 | 7 | 43 (18%) |
| On a target wheeled robot | 1 | 17 | 12 | 1 | 6 | 37 (15%) |
| *Assessment and resources* | | | | | | |
| Integrity or failure risk | 3 | 1 | 0 | 1 | 9 | 14 (6%) |
| Computation placement | 45 | 31 | 18 | 33 | 28 | 155 (64%) |
| *Platform group* | | | | | | |
| Target wheeled robot | 30 | 50 | 30 | 32 | 30 | 172 (70%) |
| Road vehicle (R) | 7 | 1 | 0 | 0 | 4 | 12 (5%) |
| Other ground robot (G) | 1 | 4 | 5 | 3 | 0 | 13 (5%) |
| Non-ground or none (N) | 29 | 1 | 1 | 10 | 6 | 47 (19%) |

Counts are per publication. A same-experiment row counts a publication only when the condition and the task outcome come from one experiment. Percentages refer to all 244 publications. Table S9 gives column percentages.

### B. Evidence Across Navigation Functions

The counts in Table V describe the selected sample rather than the field, and assessment depth varies as disclosed in the supplementary material. Three patterns are nevertheless large enough to qualify the comparisons of Sections IV–VIII. First, evidence for localization and mapping stops at the estimate, the natural endpoint of an estimation study, since all 67 studies report component measures and 61 name a degradation condition, whereas only 1 reports a task outcome. Planning and semantic studies show the opposite profile, with task outcomes in 44 of 56 and 33 of 36 studies. Second, 81 publications (33%) report a task outcome under a named degradation condition in the same experiment, the combination that ties a degraded condition to executed behavior. The share rises from 1% in localization, against 91% with a named condition and component measures, through 24% in multi-robot coordination, 40% in fault management, and 47% in semantic navigation to 64% in planning and control. Of these publications, 27 rest on a condition injected or varied against a comparator and 43 on a physical robot. The other 54 lack such a comparator, some resting on one physical run per method, as in the LRAE field test [126] and the outdoor route of VL-TGS [179]. Third, the sample contains only 14 publications with integrity or failure-risk assessment and 89 publications that do not report computation placement.

These shares are statistics of the purposively selected sample that depend on definitions and subsets (Table S11 in the supplementary material). The headline share falls to 5% when the condition must be varied against a nominal level, becomes 34% when the two attributes may come from different experiments, and was 25% under the card-level coding of an earlier draft. It is 30%, 38%, and 30% among full, earlier-extraction, and targeted readings, and 40% among target wheeled robots against 17% among other carriers. The share stays between 32% and 35% when any one selection stage is removed, at 34% in the quota stages against 32% elsewhere, and between 23% and 46% across year bands. In every stage, depth, and quota subset, localization keeps the lowest or a tied lowest share and planning the highest.

### C. Execution and Reference Dependence

Recorded measurements preserve sensor artifacts without testing feedback between the estimated state and future observations, whereas closed-loop evaluation lets the evaluated output change the commands and thus the subsequent observations. External pose, offboard processing, teleoperated prior mapping, and staged scenes define the conditions of that feedback, as when slip-aware fusion runs without a controller consuming its estimates [30] and LiDAR-based barrier control runs under external pose and offboard computation [43].

References introduce their own dependence when they share sensors or processing with the evaluated method, as in campus line localization [243], kinematic diagnosis with a substitute position [235], and orchard map merging scored against its own mapping backbone [205]. Reference quality also varies within datasets, through the mix of smoothed RTK, offline multisensor SLAM, and selected laser-tracker checks in M2UD [2], the untimestamped HK-MEMS bus-route reference [49], and scene-dependent reference construction in GEODE [47]. Cooperative range-only SLAM runs offline on joystick-recorded bags [214], whereas hierarchical map transmission reports an estimated risk field from dataset replay rather than measured contacts [204]. Controlled variables must also match the claimed mechanism. Varying terrain geometry tests plane-fitting allocation [198], whereas a test of pose-aware terrain decisions must vary pose uncertainty. Similarly, communication studies rank methods under ideal simulated links in URGE [197] and replay benchmark factors under modeled connectivity in riMESA [207], two settings that do not measure how physical link loss changes the ranking.

### D. Guarantees, Rates, and Denominators

A formal safety result applies to a specified event, probability space, horizon, and set of assumptions. A scenario-planning certificate concerns the risk of an accepted plan under a predicted obstacle distribution [146], while environment-level perception calibration depends on representative calibration environments and known state [147]. A barrier proof may assume feasible inputs for (5) that saturation violates [43]. Spatiotemporal-tube guarantees likewise assume bounded disturbances and terminal separation without explicit input limits [154]. None of these guarantees is an empirical closed-loop collision frequency, which must be estimated from executed trials.

Empirical rates require a denominator and an independent trial unit. Across ten homes, the assistive-robot study records 1539 service requests, of which 1130 succeeded, 177 failed, and 232 were not performed [3]. Its 86.46% success rate uses the 1307 attempted interventions as denominator and is therefore not the fraction of requests served. Eligibility and censoring change the represented population. Some studies remove scenarios before reporting, as when the CENTAURO study discards infeasible or slowly planned scenarios [37]. Others remove methods or runs after a failure, as cross-modal teach-and-repeat excludes methods that fail on the first route from later routes [103], an outdoor visual-inertial benchmark averages error only over surviving runs [266], and multi-session radar mapping omits failed baselines from its averages [88]. A trial repeated after failed relocalization [31] differs from recovery within the original attempt. Completion after temporal-logic relaxation [251] must likewise name the requirements that were changed. Safety and completion must also be reported together, since refusing every action reduces collisions while preventing useful work.

TABLE VI
REPORTING CHECKLIST FOR A JOINT EVALUATION OF DETECTION AND RESPONSE.

| Element | What to report |
|---|---|
| Fault model | Source, channel, and channels assumed trustworthy |
| Onset, duration | Onset time, duration, and recovery criterion |
| Severity | Tested levels, including a nominal level |
| Trial unit | Run, route, or mission, with independent repeats |
| Denominator | Attempts, censored or excluded runs, and reasons |
| Reference | Source, coverage, and dependence on tested sensors |
| Alarms | Threshold, alert limit $A_t$ in (6), false and missed alarms |
| Event times | First alarm at the supervisor and response onset |
| Delay | Detection-to-action delay in each trial |
| Computation | Placement, sensing age, and end-to-end latency |
| Controls | Nominal stack, simple response, and no-fault runs |
| Response | Executed rejection, correction, reset, or maneuver |
| Outcomes | Completion, contact, interventions, path and time cost, resource use, and component error, with intervals |

### E. Timing, Placement, and Comparisons

Timing should cover sensing age, transport, perception, estimation, planning, supervision, command delivery, and actuator response, which processor identity and average module runtime do not establish. Fast and slow monitors [245], cached reasoning [250], slow VLM planning [181], and change detection at 2.74 Hz on an embedded GPU [87] each need a local controller that stays valid between outputs. DMSA processes data at 1.5–4 times the recording duration and distinguishes unsupported sensor adaptations from failed runs [267]. Memory-event-triggered MPC counts updates on an industrial computer rather than measuring onboard execution [155]. A study of LiDAR–inertial non-reproducibility repeats each experiment 30 times after controlling subscribers and buffering [268], showing that repeated playback characterizes implementation variability rather than variation across missions.

### F. Joint Evaluation of Detection and Response

A joint evaluation links degradation onset to an internal trigger, an executed response, and a task outcome within one mission, as Fig. 3 shows. Table VI lists what such an evaluation should specify, log, and report, including matched controls that isolate the response mechanism and reveal the cost of unnecessary interventions.

## X. OPEN CHALLENGES

The four interfaces below are numbered as in Fig. 2 and specified with a matched comparison in Table VII. In the coded evidence, every pair of functions with one coupling source forms a candidate, ranked by a rule fixed in the public codebook before coding (commit 67872f4) on the number of

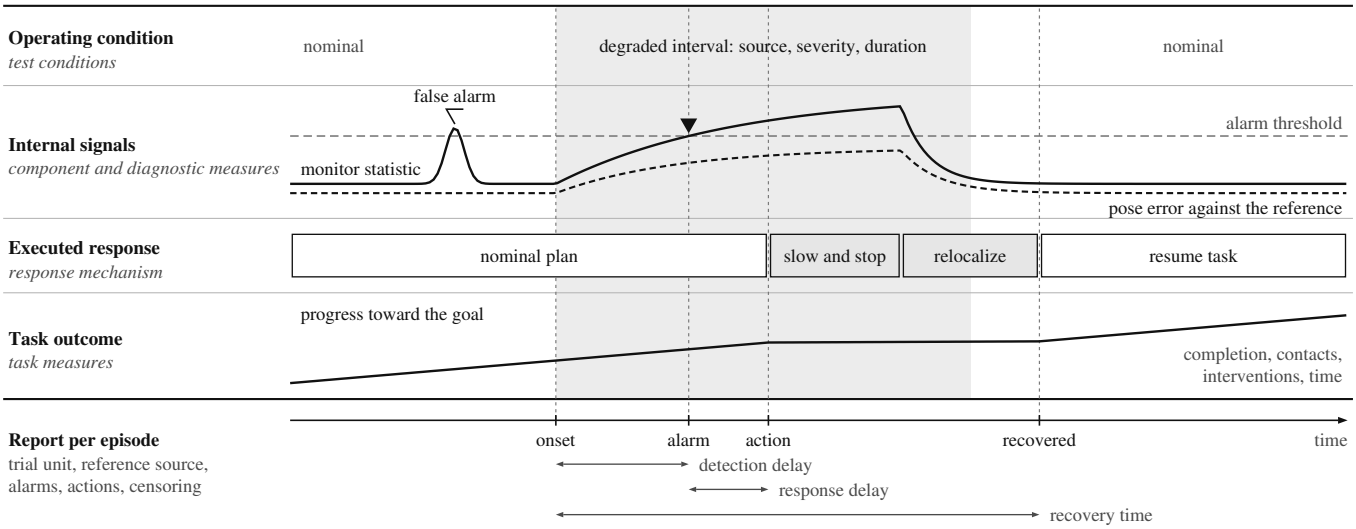



Fig. 3. Joint evaluation of detection and response in one navigation episode. The shaded band marks the degraded interval. The operating condition defines the test, internal signals carry component and diagnostic measures, and the executed response and task outcome carry task measures. Detection delay, response delay, and recovery time follow from the synchronized events. A threshold crossing while the pose error stays below the alert limit counts as a false alarm, and an error above the alert limit without a crossing counts as a missed detection.

TABLE VII
OPEN INTERFACES BETWEEN NAVIGATION FUNCTIONS, THE DEPENDENCY THAT EACH LEAVES UNRESOLVED, AND A MATCHED COMPARISON THAT WOULD TEST IT. EACH INTERFACE LISTS ITS CODED WORKS AND THOSE WITH DIRECT JOINT WHEELED EVIDENCE.

| Interface | Unresolved dependency | Matched comparison | Primary outcomes |
|---|---|---|---|
| 1. Localization-triggered recovery (Sec. X-A). 15 works, 1 direct | Which response a localization warning should select when part of the state needed to execute it is unreliable | Fixed estimator and route. Immediate stop, threshold-triggered relocalization, and an action chosen from trustworthy channels, under controlled pose bias and weak geometry, with one alarm firing while another channel misleads | Restored localization and task progress, contact and interventions, cost of unnecessary stops under nominal sensing |
| 2. Semantic reasoning with fallback (Sec. X-B). 11 works, 6 direct | How long a semantic decision stays valid, and whether wrong as well as late responses are rejected | Fixed model and controller. Periodic querying, scene-triggered querying, and cached decisions, under delayed responses, changed candidate paths, injected incorrect responses, and varied obstacle-track quality | Completion, contact, and fallback duration against energy and query cost, with the age of team assignments |
| 3. Pose uncertainty in terrain decisions (Sec. X-C). 6 works, 1 direct | How pose uncertainty changes the probability that the wheel footprint rests on traversable support | Fixed terrain map and chassis. A calibrated pose distribution propagated into footprint clearance or speed, against geometry-only planning and fixed inflation, on narrow supports and slopes with an independent reference | Contact and interventions against completion and travel time, with aborted risky attempts counted as outcomes |
| 4. Accepting shared data after reconnection (Sec. X-D). 17 works, 0 direct | Which returning information to accept when it may be stale, correlated, or biased | Moving ground team after a controlled partition. Immediate fusion, age-based rejection, and correlation- or residual-aware acceptance against a deterministic merge, with delay and landmark bias varied separately | Fault propagation and shared-state divergence, duplicated coverage, completion, and interventions |

its works without direct joint wheeled evidence, that is, without a named condition of any source and a physical task outcome in one experiment on a target wheeled robot (Section S8 of the supplementary material). At equal direct evidence, counting works ranks a candidate with more works above one with fewer, with candidate sizes shaped by the selection stages. Among the 32 candidates with at least three works, each interface takes the rank of its best candidate, 4 for Interface 1 and 11 for Interface 3 through perceptual degradation, 10 for Interface 2 through learned inference error (26 through computation limits), and 1 for Interface 4 through communication loss. Without the integration-questions stage, which was acquired for them, they rank 4, 8, 9, and 1. Ranked instead by the share of works without direct evidence, they rank 5, 26, 18, and 12, below the actuator-fault candidate named in Section X-E. The interfaces therefore remain an agenda proposed by this survey rather than the four highest-ranked candidates. Five of the 6 candidates that rank above Interface 3 without being developed are merged into Interface 4 (team communication, shared-state faults, and teammate perception) or Interface 2 (moving agents in semantic choices), with the sixth named in Section X-E.

### A. Localization-Triggered Navigation Recovery

Localization monitoring and executed recovery are both available, although the reviewed systems rarely connect them through the same trigger. Monitors cover pose-error protection levels [243], modality switching [59], landmark repair on road-vehicle data [225], and fault-tolerant tracking of specified channels [241]. Executed responses include stop–reposition–resume after optimizer failure [248], stopping and

relocalization after visual tracking loss [31], behavior-tree recovery on CENTAURO [37], view planning against tracking loss [28], and interoceptive wheel–inertial odometry whose drift grows without external correction [72]. Among the 15 coded works of this interface, only the cross-modal teach-and-repeat study gives direct joint wheeled evidence [103], whereas the integrity analyses rest on road-vehicle data [45], [46]. The open question is how a localization warning should select slowing, stopping, relocalization, or repositioning when part of the state needed to execute that action is itself unreliable. In automated driving, the corresponding response is a minimum risk maneuver toward a minimum risk condition [25], whereas a wheeled robot can also relocalize or reposition within its drivetrain limits (Table II).

### *B. Resource-Constrained Semantic Reasoning and Fallback*

Wheeled systems already combine semantic trajectory selection with geometric planning [179], [180], group reasoning with constrained local control [185], and embodiment-specific re-ranking [181], while perception calibration on a legged robot [147], runtime monitoring [245], cached reasoning with fallback on a quadrotor [250], and distilled encoders [186] address reliability and response time. Language-derived costmap layers [183] and relocation-aware object search [161] move semantic errors into costs or search order, where geometry can check them. This interface has the most direct wheeled evidence of the four, 6 of its 11 coded works, although none of them names a computation limit or response delay among its conditions. The remaining interface concerns the lifetime of a semantic decision, since an old destination can remain valid while an old pedestrian detour becomes unusable. A delayed team assignment, as in Co-NavGPT with perception and planning on a workstation [269], can likewise remain syntactically valid after the shared map or the allocation has changed. HAM-Nav and Mobility VLA depend on sketch or tour coverage, while VLM-GroNav retains GPS and cloud dependencies [170], [171], [189]. SocialNav-SUB indicates that incorrect reasoning, and not only unavailable or delayed responses, requires explicit testing [188].

### *C. Pose Uncertainty in Wheeled Terrain Decisions*

Recent terrain decisions increasingly combine geometry with localization, map, or contact uncertainty, through localizability-aware traversability [123], terrain-aware Ackermann allocation [198], uncertainty-weighted mapping on an aquatic explorer [108], uncertain elevation and terrain classes [106], [107], novelty and speed distributions [118], and angle-dependent risk [117]. Terrain-adaptive stability bounds [121] and implicit traversability fields [122] shape executed speed and route on small skid-steer robots, whose hardware failures are attributed to wheel slip, wet-ground traction, and sensor occlusion rather than to pose error. STEP comes closest to this question among these planners, since localization error enters its collision risk through the elevation-map variance, although its simulation assumes no localization error and its field runs on a Husky and a Spot quadruped report no failure rates [119]. EVORA instead models traction uncertainty under motion-capture pose indoors [120]. These terrain planners give direct wheeled evidence for terrain risk without coding localization as a function. The single direct wheeled study, OPEN, completes a GPS-disabled campus delivery with no pose-uncertainty term in its traversability cost [167], which leaves this question with extrapolated evidence only. The open question is how the pose uncertainty of (4), beyond its share of map variance, alters the probability that a wheel-supported footprint lies on the estimated traversable surface under incomplete observation, and whether propagating it changes executed outcomes.

### *D. Reconnection and Corrupted Shared Information*

Reconnection mechanisms exist for particular shared quantities, such as target estimates fused after a link returns in failure-aware aerial tracking [226], cooperative pose fusion [104], and correlation-aware landmark repair, a candidate estimator-level mechanism rather than a validated reconnection protocol [225]. Reception-aware control [53], sparse position sharing [200], and rendezvous co-optimized with allocation [219] change connectivity or message requirements. Asynchronous consensus tolerates dropped exchanges and outliers [207], [208], whereas a deterministic regional merge tolerates delay without testing corrupted content [197]. Neither outlier rejection of inter-robot loop closures [210], [211] nor store-and-forward sharing in the Subterranean Challenge teams [222]–[224] isolates the effect of stale or biased batches. Sparse loop exchange, lossy keyframe coding, layered transmission, and planned check-ins reduce traffic [204], [212], [227], [270] without establishing safe acceptance after reconnection. None of the 17 coded works of this interface gives direct joint wheeled evidence. The 12 target wheeled works among them name conditions only in recorded data, replay, online estimation, or simulation, if at all, which leaves this interface on extrapolated evidence. The open interface is selective acceptance beyond outlier rejection, since receiving a packet establishes neither frame compatibility nor the validity of its map or pose content. Concrete prerequisites to violate one at a time include stable metric odometry in foundation-model loop fusion [213], continuous range visibility in cooperative initialization [214], orchard-specific overlap priors [205], and stable front-end odometry in asynchronous collaborative LiDAR SLAM [208].

### *E. Broader Challenges*

Transfer across terrain, supervision, platforms, and simulators [32], [122], [137], [247] calls for tests on withheld conditions. Computation and energy choices [250], [257], [258], [261] require response-time distributions under concurrent loads. Map maintenance, long deployments, and continual learning [3], [81], [84], [87], [103], [186], [263], [264] must be related to revisits, availability, and human escalation, which 10 coded works note as human assistance outside the ranked function pairs. Assurance methods [46], [146]–[148] need online monitors of their own assumptions. Embodiment-specific affordances and failure-aware team policies [181], [256] call

for cross-platform tests of clearance, curvature, and traction. Among the couplings ranked above Interface 3, actuator faults accommodated inside one tracking controller [237], [239], [241] still lack tests of a planning or supervision response.

## XI. Conclusion

This survey compared methods for wheeled autonomy through the navigation assumptions affected by degradation, the information or actions changed by each response, and the evidence for their effectiveness. Selective updates, independent motion sources, model adaptation, and map revision depend on different operating conditions, namely surviving geometry, credible supervision, valid correspondences, and consistent correlations.

The reviewed results also reveal conditional trade-offs, since more observable exploration routes can cost additional travel and computation, risk-constrained planning trades conservatism against progress, and recovery helps when its predicted failure matches the obstacle to task execution. Semantic ranking remains limited by the available candidates and by the capabilities of the executing chassis, whereas compact or asynchronous communication preserves only the information encoded in the shared representation.

The central requirement for further evaluation is to connect degradation onset, an internal trigger, and an executed response within the same mission, as the four interfaces of Section X illustrate. The coded sample shows how unevenly this connection is made, since 33% of the technical publications in the purposively selected sample report a task outcome under a named degradation condition in the same experiment, from 1% of localization and mapping studies to 64% of planning and control studies. Only 11% obtain this combination under a condition injected or varied against a comparator. The headline share is 40% among the 172 publications evaluated on target wheeled robots or their data, while only 43 publications obtain the combination on a physical robot. Comparisons that retain the trial unit, reference dependence, timing, and failure conditions can establish when an integrated response improves both task performance and safety. The heterogeneous, purposively selected evidence supports these mechanism-level conclusions, whereas broader rankings require matched protocols and more uniform coverage.

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